

Digital Electronics ...

→ Signal

A signal is a function that represents variation of a physical quantity wrt any parameter

Usually signal is variation of electrical quantity (I or V) wrt time

→ Analog signal

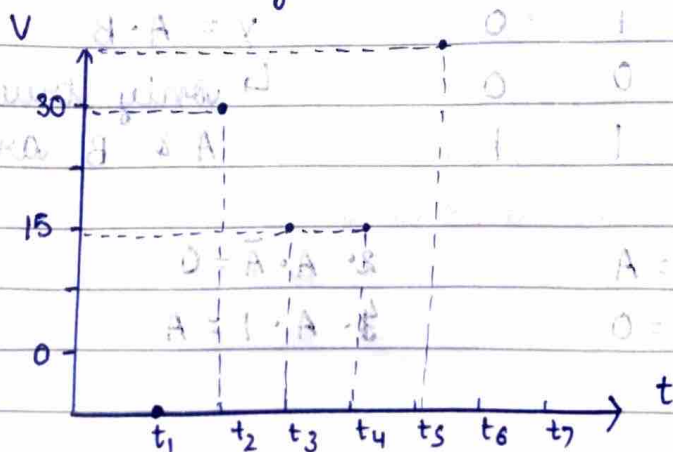
signals which can take any value within a given limit

→ Discrete Time signal

signal which is defined for discrete time intervals
It's a subset of analog signal

→ Digital signal

In digital signals, we discretise both time and the magnitude. We divide magnitude axis into fixed no of levels and signal can take values equal to these levels only.



→ Need of digital signals

All real life signals are analog. Digital signal is used in communication process to minimize effect of noise (unwanted signals)

↳ when value of noise is low

→ Advantage of digital signals

• Noise immunity

• Uses less bandwidth

• Encrypted

• Greater efficiency for long distance transmission

→ Boolean Algebra

It's set of rules used to simplify the logic without changing functionality. It's used when variables are less

→ Complement

A complement = $\bar{A}$ or A' or not A

$$\bar{0} = 1 \quad \text{and} \quad \bar{1} = 0$$

$$\underline{(A')' = A}$$

→ AND

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



0 → false

1 → true

$$Y = A \cdot B$$

↳ only true when both A & B are true

1. $A \cdot A = A$

2. $A \cdot \bar{A} = 0$

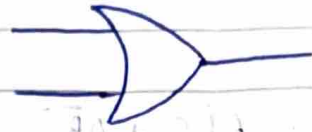
3. $A \cdot 0 = 0$

3. $A \cdot 1 = A$

→ OR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

$$Y = A + B$$



Y is true when either A or B is true

1. $A + A = 0$

2. $A + 0 = A$

3. $\bar{A} + A = 1$

4. $A + 1 = 1$

→ Distributive law

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

$$\Rightarrow \underline{A + \bar{A}B = A + B}$$

→ Commutative law

$$AB = BA$$

$$A + B = B + A$$

(NOT) > (AND) > (OR)

Order of priority

→ Associative law

$$(AB)C = A(BC)$$

→ Demorgan's Law

$$\overline{(A+B)} = \bar{A} \cdot \bar{B}$$

$$\overline{(AB)} = \bar{A} + \bar{B}$$

$$Q \quad Y = \bar{B}A\bar{C} + \bar{B}A\bar{C} + \bar{B}A\bar{C} + \bar{B}A\bar{C}$$

$$B\bar{C}(A+1) + \bar{B}A\bar{C} = B\bar{C} + \bar{B}A\bar{C}$$

$$\bar{C}(B + \bar{B}A) = \bar{C}((B + \bar{B}) \cdot (B + A))$$

$$\bar{C}(B + A) = B\bar{C} + A\bar{C}$$

$$Q \quad AB + A\bar{B}$$

$$= A(B + \bar{B})$$

$$= A$$

$$Q \quad AB + A\bar{B}C + A\bar{B}\bar{C}$$

$$= AB + A\bar{B}(C + \bar{C})$$

$$= AB + A\bar{B} = A(B + \bar{B})$$

$$= A$$

$$Q \quad (A+B+C)(A+\bar{B}+C)(A+B+\bar{C})$$

$$= \text{Let } A+B = X$$

$$= (X+C)(A+\bar{B}+C)(X+\bar{C})$$

$$= (X+C)(X+\bar{C})(A+\bar{B}+C)$$

$$= [X + X(C+\bar{C}) + 0](A+\bar{B}+C)$$

$$= X(A+\bar{B}+C)$$

$$= (A+B)(A+\bar{B}+C)$$

$$= A + B(\bar{B}+C) + B(A+C) + B\bar{B}$$

$$= A + BC + AB + BC$$

$$= A(B+1) + BC$$

$$= A + BC$$

$$Q \quad (A+B)(A+\bar{B})(\bar{A}+B)(\bar{A}+\bar{B})$$

$$(A+B)(\bar{A}+\bar{B})(A+\bar{B})(\bar{A}+B)$$

$$(A+B)(\bar{A}+\bar{B})(A+\bar{B})(\bar{A}+B)$$

$$= (A\bar{B} + B\bar{A})(AB + \bar{A}\bar{B})$$

$$= (A\bar{B} + B\bar{A})(AB + \bar{A}\bar{B})$$

$$= (A + B\bar{B})(\bar{A} + B\bar{B})$$

$$= A\bar{A}((A+B) \cdot (B+1)) = (A\bar{A} + 1) \cdot 0$$

$$= 0$$

→ Redundancy Theorem / Consensus Theorem
conditions to use :-

- 3 variables
- each variable repeated twice
- one variable is complemented
- Take complemented variable & omit term in which complemented variable

If $Y = AB + \bar{A}C + BC$

Then

$Y = AB + \bar{A}C$

also

PROOF

eg → $Y = AB + \bar{A}C + BC \cdot 1$ $\bar{A} + A$ isn't there

$= AB + \bar{A}C + BC(A + \bar{A})$

$= AB + \bar{A}C + ABC + \bar{A}BC$

$= AB(1+C) + \bar{A}C(1+B)$

$= AB + \bar{A}C$

Q $AB + B\bar{C} + AC$
 $= B\bar{C} + AC$

Q $A\bar{B} + BC + AC$
 $= A\bar{B} + BC$

Q $(A+B)(\bar{A}C)(B+C)$
 $(A+B)(\bar{A}+C)$

Q $(A+B)(\bar{B}+C)(A+C)$
 $(A+B)(\bar{B}+C)$

Q $\bar{A}\bar{B} + A\bar{C} + \bar{B}\bar{C}$
 $\bar{A}\bar{B} + A\bar{C}$

Here all were 'complemented' but A wasn't so omit term not having A

min terms → used by and-ing 2 or more variables

→ Sum of Products (SOP) Form

	A	B	C	F
m ₀ 0	0	0	0	0
m ₁ 1	0	0	1	0
m ₂ 2	0	1	0	1
m ₃ 3	0	1	1	0
m ₄ 4	1	0	0	1
m ₅ 5	1	0	1	1
m ₆ 6	1	1	0	1
m ₇ 7	1	1	1	1

Decimal equivalent

(SOP) Form

$\bar{A}\bar{B}C + A\bar{B}\bar{C}$
 $+ A\bar{B}C + A\bar{B}C$
 $+ ABC$

used for high outputs (1's)

SOP + POS

$F = \sum m(2, 4, 5, 6, 7)$
(A, B, C)

Standard / Canonical SOP form

Now,

$$F = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}\bar{B}\bar{C} + A\bar{B}(C + \bar{C}) + AB(C + \bar{C})$$

$$= \bar{A}\bar{B}\bar{C} + A\bar{B} + AB$$

$$= \bar{A}\bar{B}\bar{C} + A(\bar{B} + B)$$

$$= \bar{A}\bar{B}\bar{C} + A$$

$$= (A + \bar{A}) \cdot (A + \bar{B}\bar{C})$$

$$F = A + \bar{B}\bar{C}$$

minimal SOP form

Q Minimise SOP

A B Y

0 0 0

0 1 1

1 0 0

1 1 1

$$F = \bar{A}\bar{B} + AB$$

$$= B(A + \bar{A})$$

$$= B$$

Q $Y(A, B) = \sum m(0, 2, 3)$

A B Y

m₀ 0 0 1m₁ 0 1 0m₂ 1 0 1m₃ 1 1 1

$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$= \bar{A}\bar{B} + A(\bar{B} + B)$$

$$= \bar{A}\bar{B} + A$$

$$= (A + \bar{A})(A + \bar{B})$$

$$= A + \bar{B}$$

$$= A + \bar{B}$$

$$= A + \bar{B}$$

$$= A + \bar{B}$$

$$= A + \bar{B}$$

$$= A + \bar{B}$$

→ Product of Sum (POS Form) → used for low outputs (0's)

	A	B	C	Y
M ₀	0	0	0	0
M ₁	0	0	1	0
M ₂	0	1	0	1
M ₃	0	1	1	0
M ₄	1	0	0	1
M ₅	1	0	1	1
M ₆	1	1	0	1
M ₇	1	1	1	1

SOP	POS
0 → $\bar{A}$	0 → A
1 → A	1 → $\bar{A}$

$Y = \prod (M_0, M_1, M_3)$

$Y = (A+B+C)(A+B+\bar{C})(A+\bar{B}+\bar{C}) \rightarrow \text{Normal form}$

$\bar{Y} = (\bar{A}\bar{B}\bar{C}) + (\bar{A}\bar{B}C) + (\bar{A}B\bar{C})$

$\bar{Y} = \bar{A}\bar{B}(C+\bar{C}) + \bar{A}B\bar{C}$

$\bar{Y} = \bar{A}\bar{B} + \bar{A}B\bar{C} = \bar{A}(\bar{B} + B\bar{C}) = \bar{A}(\bar{B} + C)$

$\Rightarrow Y = A + (\bar{B} + C)\bar{A} = A + \bar{B}\bar{C} \rightarrow \text{minimal POS form}$

Q Minimise POS

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

$\Rightarrow Y = (A+\bar{B})(\bar{A}+\bar{B})$

$\Rightarrow Y = \bar{A}\bar{B} + \bar{A}\bar{B} + \bar{B}$

$\bar{A} = \bar{A}\bar{B} + \bar{A}B$

$\bar{B} = \bar{A}\bar{B} + A\bar{B}$

$\bar{B} = \bar{A}\bar{B} + A\bar{B}$

$(\bar{A}\bar{B} + \bar{A}B) + (\bar{A}\bar{B} + A\bar{B}) = \bar{A}\bar{B} + \bar{A}B + \bar{A}\bar{B} + A\bar{B}$

$\bar{A}\bar{B} + \bar{A}B + \bar{A}\bar{B} + A\bar{B}$

$\bar{A}\bar{B} + \bar{A}B$

$\bar{A}\bar{B} + \bar{A}B$

Q. $Y(A, B, C) = \sum m(0, 2, 3, 6, 7)$ max.

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

→ POS $Y = (0 + \bar{A} + \bar{B}) (0 + \bar{A} + B) (0 + A + B) = Y$

$$Y = (A + B + \bar{C}) (\bar{A} + B + C) (\bar{A} + B + \bar{C})$$

$$\bar{Y} = \bar{A} \bar{B} C + A \bar{B} \bar{C} + A \bar{B} C$$

$$(0 + \bar{Y}) = \bar{A} \bar{B} C + A \bar{B} \bar{C} = \bar{B} (A + \bar{A} C)$$

$$\bar{Y} = \bar{B} (A + C)$$

$$Y = B + (A + C) = AB + \bar{A}C + A + \bar{A}C$$

$$(\bar{A} + B)(\bar{C} + B)$$

→ SOP

$$Y = \bar{A} \bar{B} \bar{C} + \bar{A} B \bar{C} + \bar{A} B C + A \bar{B} \bar{C} + A B C$$

$$= \bar{A} \bar{C} (\bar{B} + B) + \bar{A} B C + A B (C + \bar{C})$$

$$= \bar{A} \bar{C} + A B + \bar{A} B C$$

$$= (\bar{A} \bar{C}) + A B (A + \bar{A} C)$$

$$= \bar{A} \bar{C} + A B (A + C) =$$

$$= \bar{A} \bar{C} + A B + A B C$$

$$(A + \bar{A}) (A + \bar{A} C) =$$

$$= A + \bar{A} C$$

$$= A + \bar{A} C$$

on solving $B + \bar{A} \bar{C}$

Q Consider a logic ckt with 3 inputs A, B, C. Out put Y is 1 (high) for:

i) B & C true

ii) A & C false

iii) A, B & C true

iv) A, B, C false

Minimise Y

$$Y = BC + \bar{A}\bar{C} + ABC + \bar{A}\bar{B}\bar{C}$$

$$= BC(1+A) + \bar{A}\bar{C}(1+B)$$

$$= BC + \bar{A}\bar{C}$$

Q Logic ckt have 3 inputs A, B & C. Output F is high when majority inputs are logic 1. Minimise F

A B C F

0 0 0 0

0 0 1 0

0 1 0 0

0 1 1 1

1 0 0 0

1 0 1 1

1 1 0 1

1 1 1 1

$$F = 0\bar{A}\bar{B}C + 0\bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= C(\bar{A}\bar{B} + A\bar{B}) + AB$$

$$= \bar{A}BC + A\bar{B}C + AB$$

$$= \bar{A}BC + A(B + \bar{B}C)$$

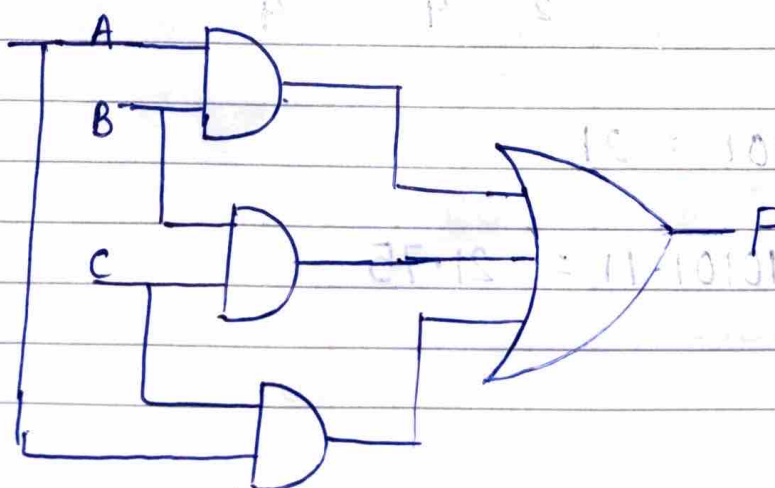
$$= \bar{A}BC + A(B + C)$$

$$= \bar{A}BC + AB + AC$$

$$= B(A + \bar{A}C) + AC$$

$$= B(A + C) + AC$$

$$\Rightarrow F = AB + AC + BC$$



→ Number system

Name	Base	(0, 1)
Binary	2	(0, 1)
Octal	8	(0, 1, 2, 3, 4, 5, 6, 7)
Decimal	10	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
Duodecimal	12	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b)
Hexadecimal	16	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b, c, d, e, f)

→ Binary system

0, 1 → Bits (smallest unit of data)

1 Nibble = 4 bits

→ 1 byte = 8 bits

1 word = 16 bits

$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 1 & \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & \end{array}$$

$$= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 16 + 0 + 4 + 0 + 1$$

$$= 21$$

$$= 10101_2$$

$$10101_2 \rightarrow 2^{-2}$$

$$10101_2 \rightarrow 2^{-1}$$

$$10101_2 \rightarrow 2^{-2}$$

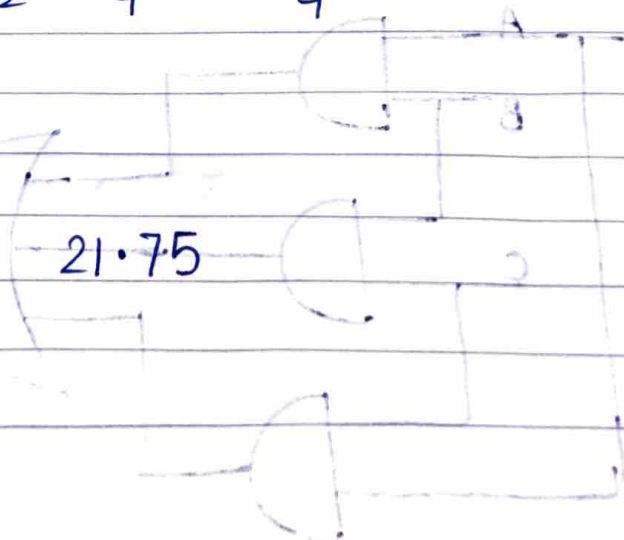
$$10101_2 = 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^0$$

$$= 16 + 4 + 1$$

$$= \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$10101 = 21$$

$$10101.11 = 21.75$$



→ MSB and LSB → least significant bit
most significant bit

eg → 10101
 $\downarrow$ MSB $\downarrow$ LSB

It's MSB because if we change it to 0, no becomes 00101 which is 5 but earlier it was 21 so it brings much larger change & hence it's MSB

→ Decimal to Binary conversion

$$13 \rightarrow 2^3 + 2^2 + 2^0$$

$$= 1101$$

write in sum of powers of 2

remainders $1101 \rightarrow$ Binary equivalents

2	13	1
2	6	0
2	3	1
	1	

• 25.625

2	25	1
2	12	0
2	6	0
2	3	1
	1	

0.625

$$0.625 \times 2 = 1.25$$

Take out int part 1 X by base

$$0.25 \times 2 = 0.5$$

$$0.50 \times 2 = 1.00$$

$$0.00 \times 2 = 0.00$$

top to bottom

$$0.625 \approx 10100$$

$$25.625 \approx 11001.10100$$

→ decimal to octal conversion

- 112

8	112	0
8	14	6
	1	

$$= 160$$

- 25.625

8	25	1
	3	31

$$0.625 \times 8 = 5.000$$

$$0.000 \times 8 = 0.000$$

$$0.50 \times 8 = 4.000$$

$$0.50 \times 8 = 4.000$$

→ decimal to Hexadecimal

- 254

16	254	14
16	15	15
	0	

$$1514$$

$$15 \rightarrow F$$

$$14 \rightarrow E$$

$$10 \rightarrow A$$

$$11 \rightarrow B$$

$$12 \rightarrow C$$

$$13 \rightarrow D$$

$$254 \rightarrow FE$$

- 25.625

16	25	9
16	1	01
	00	00

$$0.625 \times 16 = 10.000$$

$$0.000 \times 16 = 0.000$$

$$0.000 \times 16 = 0.000$$

$$0.000 \times 16 = 0.000$$

$$00101 \times 250.0$$

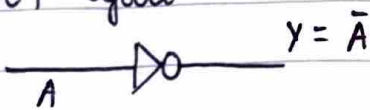
$$00101 \times 10011 \times 250.0$$

→ Logic gates

It's a physical device which performs logic operation on one or more inputs and produce single logical output

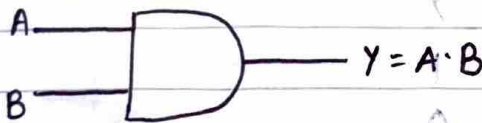
- Basic gate NOT, AND, OR
- Universal gate NAND, NOR
- Arithmetic gate XOR, XNOR

→ NOT gate

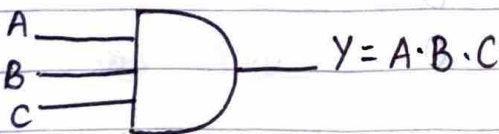


A	Y
1	0
0	1

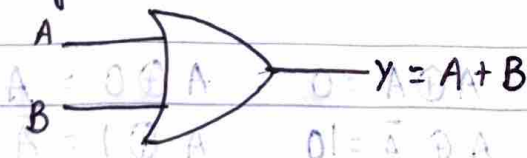
→ AND gate



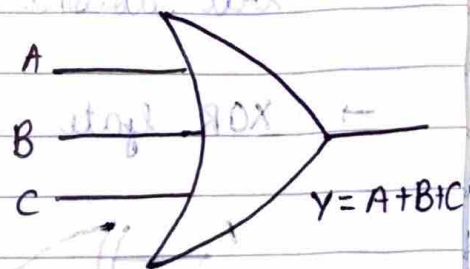
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



→ OR gate

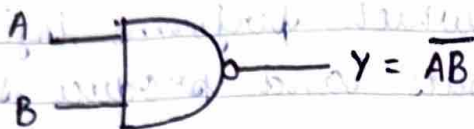


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



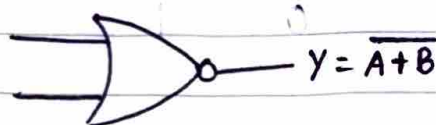
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

→ NAND gate



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

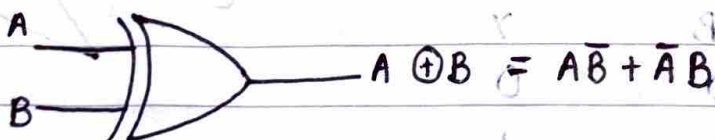
→ NOR gate



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

→ Universal gates follow commutative law but doesn't follow associative law

→ XOR gate



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\begin{aligned} A \oplus A &= 0 & A \oplus 0 &= A \\ A \oplus \overline{A} &= 1 & A \oplus 1 &= \overline{A} \end{aligned}$$

If $A \oplus B = C$

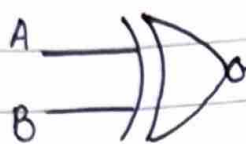
Then

$$A \oplus C = B$$

$$B \oplus C = A$$

$$A \oplus B \oplus C = 0$$

→ XNOR gate



$$Y = A \odot B = A \cdot B + \bar{A} \bar{B}$$

$$A \odot A = 1$$

$$A \odot \bar{A} = 0$$

$$A \odot 0 = \bar{A}$$

$$A \odot 1 = A$$

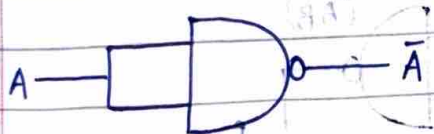
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

output = 1

when $A = B$

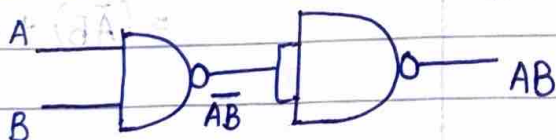
→ NAND gate as universal gates

- NAND as NOT



1 NAND

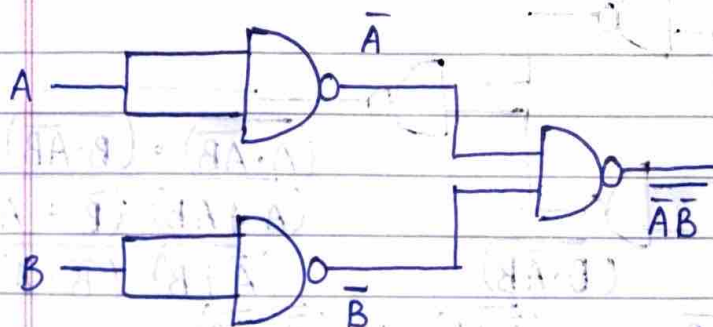
- NAND as AND



2 NAND

- NAND as OR

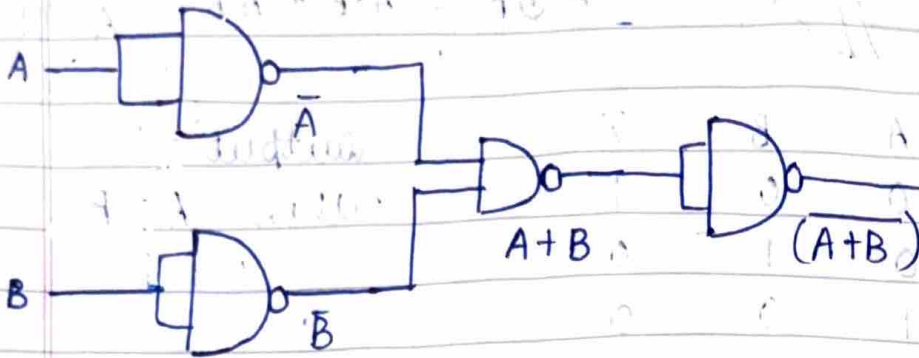
$$Y = A + B = \overline{\bar{A} \cdot \bar{B}}$$



3 NAND

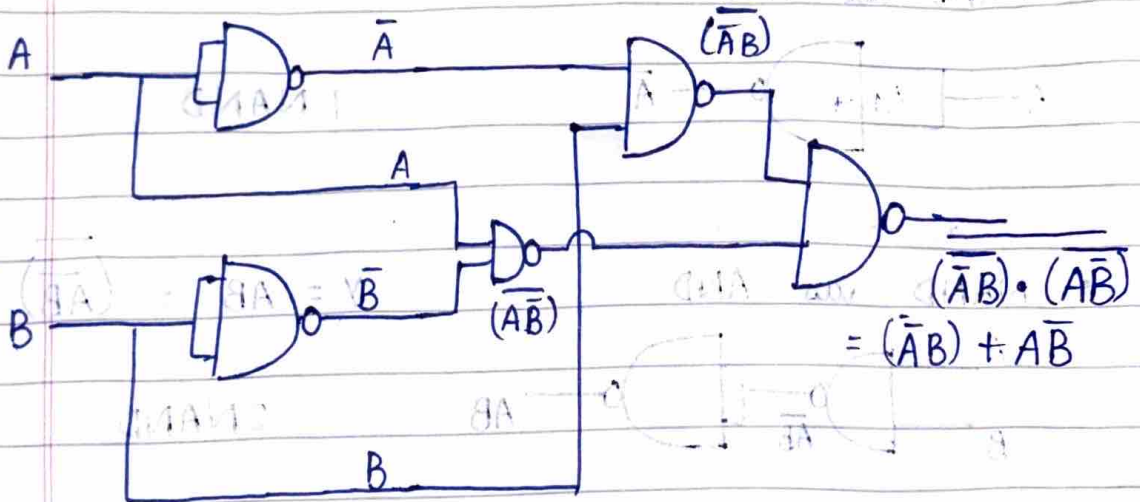
$$\bar{A} \bar{B} = A + B$$

- NAND as NOR
 $\overline{A+B}$

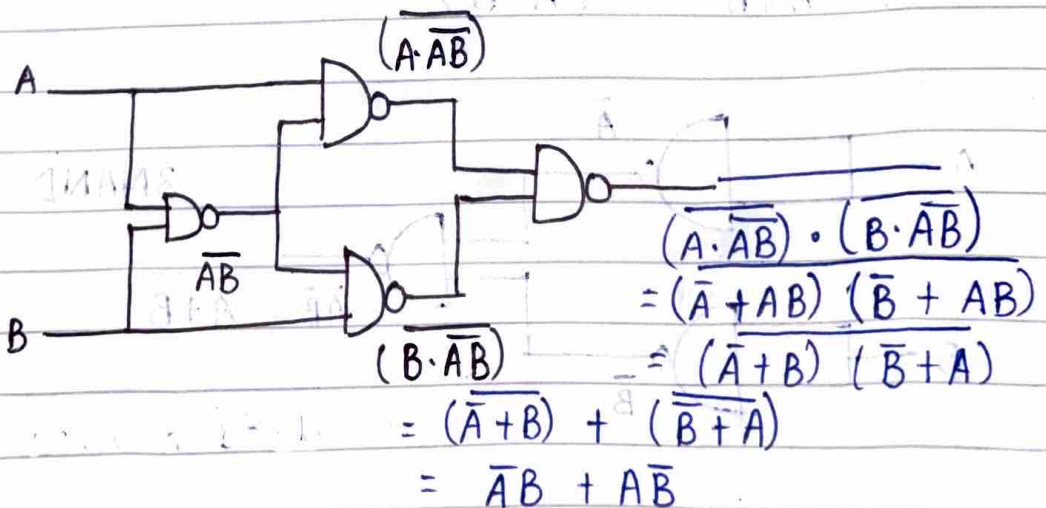


- NAND as XOR

$$Y = \bar{A}\bar{B} + A\bar{B}$$

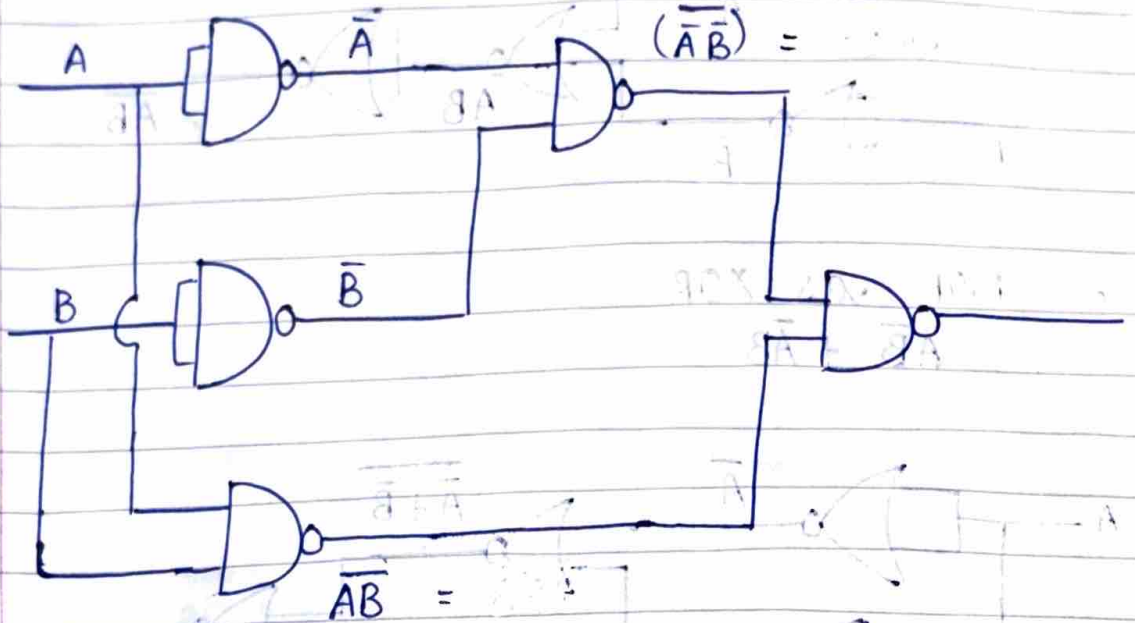


OR



- NAND as XNOR

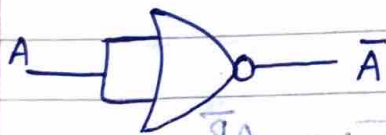
$$Y = AB + \bar{A}\bar{B}$$



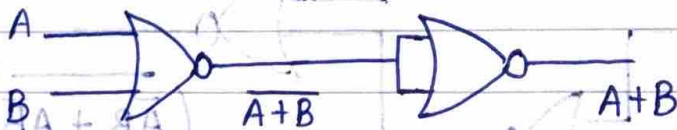
$$Y = AB + (\overline{AB})(\overline{AB}) = AB + \bar{A}\bar{B} + AB$$

- NOR as universal gate

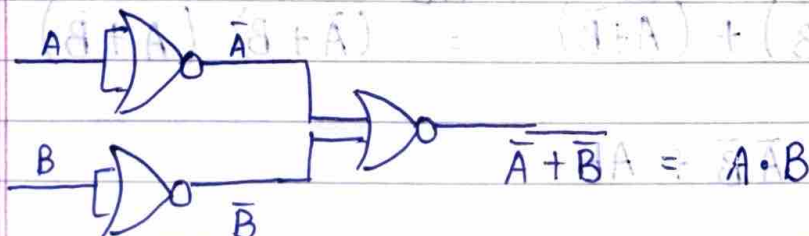
- NOR as NOT



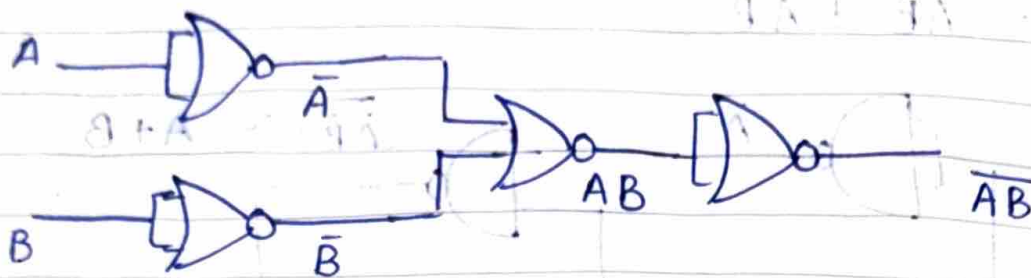
- NOR as OR



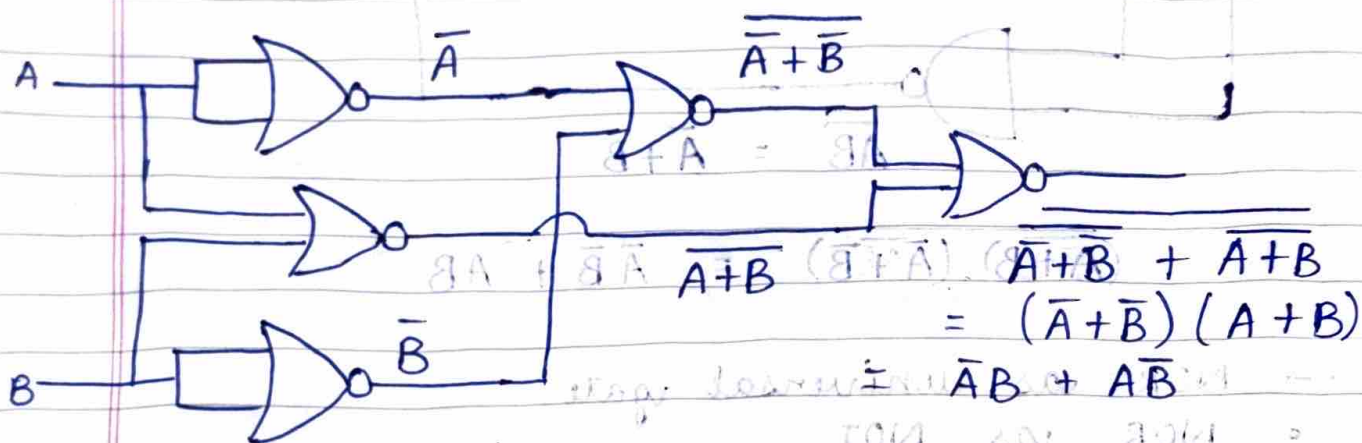
- NOR as AND



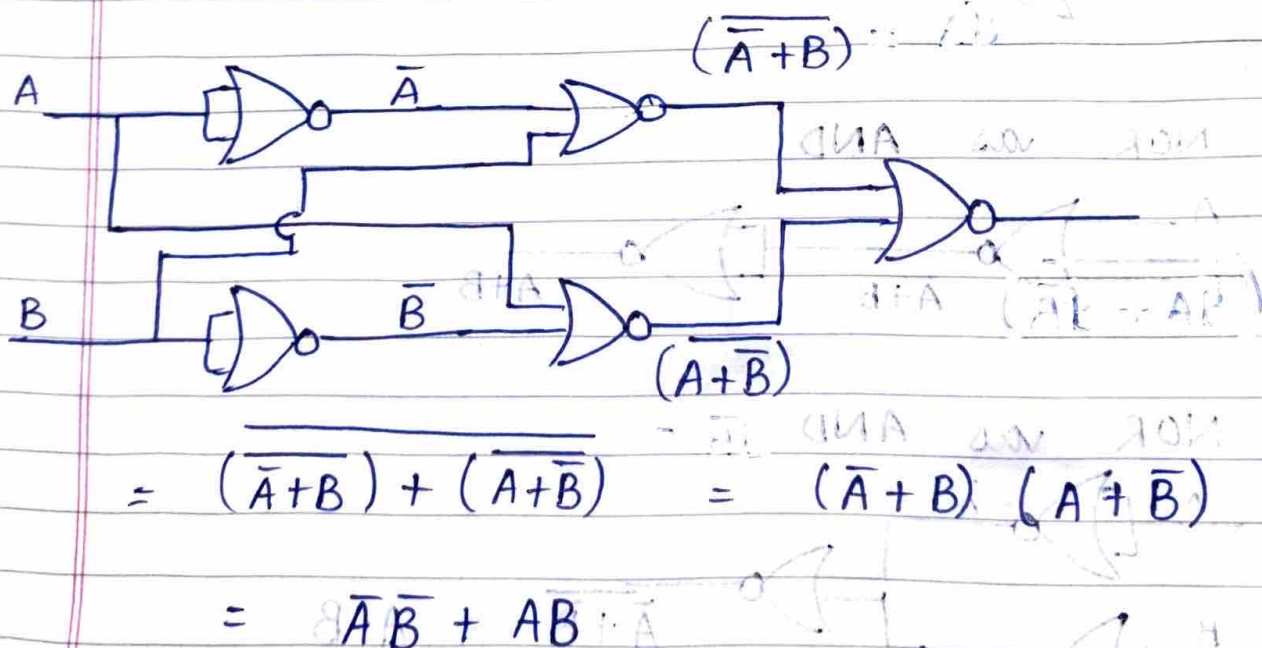
- NOR as NAND



- NOR as XOR
 $AB + \overline{A}\overline{B}$



- NOR as XNOR



→ Karnaugh Map (K-Map)

$$F = A + B\bar{C}$$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$F = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + ABC + ABC$$

$$F = \bar{A}\bar{B}\bar{C} + A\bar{B} + AB$$

$$= A + \bar{A}\bar{B}\bar{C}$$

$$= A + B\bar{C}$$

only 1 bit can be changed in adjacent cells

MSB A \ BC LSB	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	m_0	m_1	m_3	m_2
A	m_4	m_5	m_7	m_6

Gray code

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	0	0	1
A	1	1	1	1

Pairs of 1's made in powers of 2.

2, 4, 8, 16, 32, ...

$$\left. \begin{array}{l} G_1 = m(4, 5, 6, 7) = A \\ G_2 = m(2, 6) = B\bar{C} \end{array} \right\} \text{To find group we common raha}$$

$$\Rightarrow Y = A + B\bar{C}$$

Q $f(A, B, C) = \sum m(1, 3, 5, 7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1	0
A	0	1	1	0

$$= \bar{A}C + AC$$

$$= C(A + \bar{A})$$

$$= C$$

Q $f(A, B, C) = \sum m(0, 1, 2, 4, 7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1	1	0	1
A	1	0	1	0

$$= \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}BC + \bar{A}B\bar{C}$$

$$= \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC$$

$$= \bar{A}\bar{B} + \bar{C}(\bar{B} + B\bar{A}) + \bar{A}BC$$

$$= \bar{A}\bar{B} + \bar{C}(\bar{B} + \bar{A}) + \bar{A}BC$$

$$= \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C} + \bar{A}BC$$

Q $f(A, B, C) = \sum m(1, 3, 6, 7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	1	0
A	0	0	1	1

$$\bar{A}C + AB$$

Q $f(A, B, C) = \sum m(0, 1, 5, 6, 7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$BC\bar{C}$
$\bar{A}$	1	1	0	0
A	0	1	1	1

$$\bar{A}\bar{B} + \bar{B}C + AB$$

→ 4 variable K-MAP

$AB \backslash CD$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	m_0	m_1	m_3	m_2
$\bar{A}B$	m_4	m_5	m_7	m_6
AB	m_{12}	m_{13}	m_{15}	m_{14}
$A\bar{B}$	m_8	m_9	m_{11}	m_{10}

Q $f(A, B, C, D) = \sum m(0, 2, 3, 7, 11, 13, 14, 15)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	0	1	1
$\bar{A}B$	0	0	1	0
AB	0	1	1	1
$A\bar{B}$	0	0	1	0

$$\bar{A}\bar{B}\bar{D} + CD + ABD + ABC$$

Q $f(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$		1	1	
AB			1	1
$A\bar{B}$	1		1	1

$$CD + \bar{B}\bar{D} + AC + \bar{A}BD$$

$$\bar{C}\bar{D}\bar{B} + \bar{A}BD + \bar{A}\bar{B}C + ACD$$

or

$$G_1 \rightarrow m(0, 2, 8, 10)$$

$$G_2 \rightarrow m(3, 7, 15, 11)$$

$$G_3 \rightarrow m(5, 7)$$

$$G_4 \rightarrow m(10, 11, 14, 15)$$

(x, y, z, w)

Q $F = \sum m(1, 5, 7, 9, 11, 13, 15)$

	$\bar{z}\bar{w}$	$\bar{z}w$	zw	$z\bar{w}$
$\bar{x}\bar{y}$	0	1	0	0
$\bar{x}y$	0	1	1	0
xy	0	1	1	0
$x\bar{y}$	0	1	1	0

$$= xw + yw + \bar{x}\bar{z}w$$

→ Don't care case in K maps

Q $F(A, B, C) = \sum m(2, 3, 4, 5) + \sum d(6, 7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	0	1	1
A	1	1	X	X

→ K Map using max terms (group 0's)

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	0	0	1
A	1	1	1	1

output is $\bar{F}$

$$\bar{F} = \bar{A}\bar{B} + \bar{A}C$$

$$F = (A+B) \cdot (A+C)$$

POS form

$$(0, 1, 2, 3, 4, 5, 6, 7) \text{ m} \rightarrow 10$$

$$(1, 2, 3, 4, 5, 6, 7) \text{ m} \rightarrow 10$$

$$(1, 2, 3, 4, 5, 6, 7) \text{ m} \rightarrow 10$$

$$(1, 2, 3, 4, 5, 6, 7) \text{ m} \rightarrow 10$$